

### PROBLEM SHEET 3

This problem sheet is due in by **5pm** on **Wednesday 26 November 2025**. You may submit it on paper to my office (MCS 3008) or electronically on Gradescope.

**Topics covered:** Chapters 4–5.

1. **Bifurcation investigation** Consider this version of the nondimensionalised Rosenzweig–MacArthur predator–prey model,

$$\frac{du}{dt} = u \left( 1 - \frac{u}{k} \right) - \frac{auv}{1+u}, \quad (1a)$$

$$\frac{dv}{dt} = -bv + \frac{auv}{1+u}, \quad (1b)$$

where  $a$ ,  $b$  and  $k$  are positive constants.

- (a) Provide a biological interpretation of the interaction term between the two species. Can you summarise it in three words?
- (b) Show that there are two equilibria of the system,  $(u_0, v_0) = (0, 0)$  and  $(k, 0)$ , which are always permissible; and that there is a further,  $(u^*, v^*)$ , which is permissible under conditions you should state.
- (c) From now on, let  $a = 2$  and  $b = 1$ . What does this make the value of the coexistence equilibrium,  $(u^*, v^*)$ , and what are the new conditions for it to be permissible?
- (d) Determine the conditions under which each of the three equilibria is stable or unstable. In particular, you should show that the Jacobian of the system, evaluated at the coexistence equilibrium, is

$$J(u^*, v^*) = \begin{pmatrix} \frac{k-3}{2k} & -1 \\ \frac{k-1}{2k} & 0 \end{pmatrix}; \quad (2)$$

and you will find that using the determinant and trace is the best strategy for evaluating the conditions for this equilibrium's stability.

- (e) Now consider how each of the equilibria,  $(u_0, v_0)$ , depend on  $k$ . Ignoring the complete extinction equilibrium, draw a bifurcation diagram of  $u_0$  against  $k$ ; and another of  $v_0$  against  $k$ . You should remember to indicate where each of the remaining two equilibria is stable/unstable using solid/dotted lines.
- (f) Finally, indicate the point on each bifurcation diagram where a Hopf bifurcation occurs.

2. **Bacterial equilibrium** Consider a species of bacteria represented by its density (number per unit length)  $u(x, t)$  and constrained to a long thin tube of length  $L$ , which we will represent as a 1D domain,  $x \in [0, L]$ .

The bacteria move subject to diffusion alone, i.e. with no advection ( $v = 0$ ). This diffusion is determined by Fick's law, and competition for oxygen within the tube ensures the bacteria die at a rate which is proportional to the bacteria density  $u(x, t)$ , with a constant of proportionality  $\lambda$ .

At one end of the tube,  $x = 0$ , there is some chemical which kills the bacteria instantaneously. At the other end,  $x = L$ , the bacteria are flowing in with a density gradient  $\mu$ .

- (a) Explain why the bacteria should be modelled as

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2} - \lambda u, \quad (3)$$

and state the boundary conditions. You should start from the general advection–diffusion equation.

- (b) Show that the equilibrium of the system,  $u(x)$ , in the case  $\lambda = 1$  is given by

$$u(x) = \frac{\mu \alpha \sinh(x/\alpha)}{\cosh \beta}, \quad (4)$$

where you should replace  $\alpha$  and  $\beta$  with appropriate constants.

- (c) Hence show that the centre of mass of the distribution  $u$ , which is defined as

$$C_m = \frac{\int_0^L x u(x) \, dx}{\int_0^L u(x) \, dx}, \quad (5)$$

is given by

$$C_m = \frac{\gamma \cosh \beta - \alpha \sinh \beta}{\cosh \beta - \delta}, \quad (6)$$

where you should replace  $\alpha$ ,  $\beta$ ,  $\gamma$  and  $\delta$  with appropriate constants.

- (d) What happens to the centre of mass as the domain size gets significantly larger than the diffusion coefficient?